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Hsiao et al.

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(54) **ELECTRICAL CONNECTION BASE,
ELECTRICAL CONNECTION BASE
ASSEMBLY WITH ENHANCED
LOAD-BEARING AND SUPPORTIVE
ABILITY, AND ELECTRICAL CONNECTOR**

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13/635 (2013.01)

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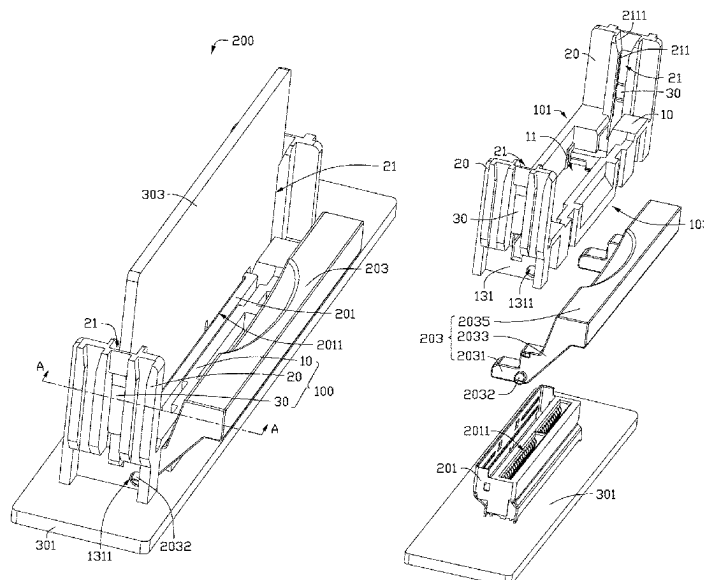
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(57) **ABSTRACT**

An electrical connection base providing support for heavy expansion cards includes a pedestal and a supporting element. A socket is defined on the pedestal. The supporting element includes installation and support portion. A through hole is defined on the installation portion, which is portion set outside the pedestal, the pedestal entering the through hole. The support portion has a first slot, and the first slot runs through the end of the support portion away from the installation portion. The portion of the expansion card or other element plugged into the socket and outside the pedestal is inserted into the first slot. The groove wall of the first slot is in contact with the element, the support portion thereby supports the card or element and prevents strain and deformation.

20 Claims, 10 Drawing Sheets



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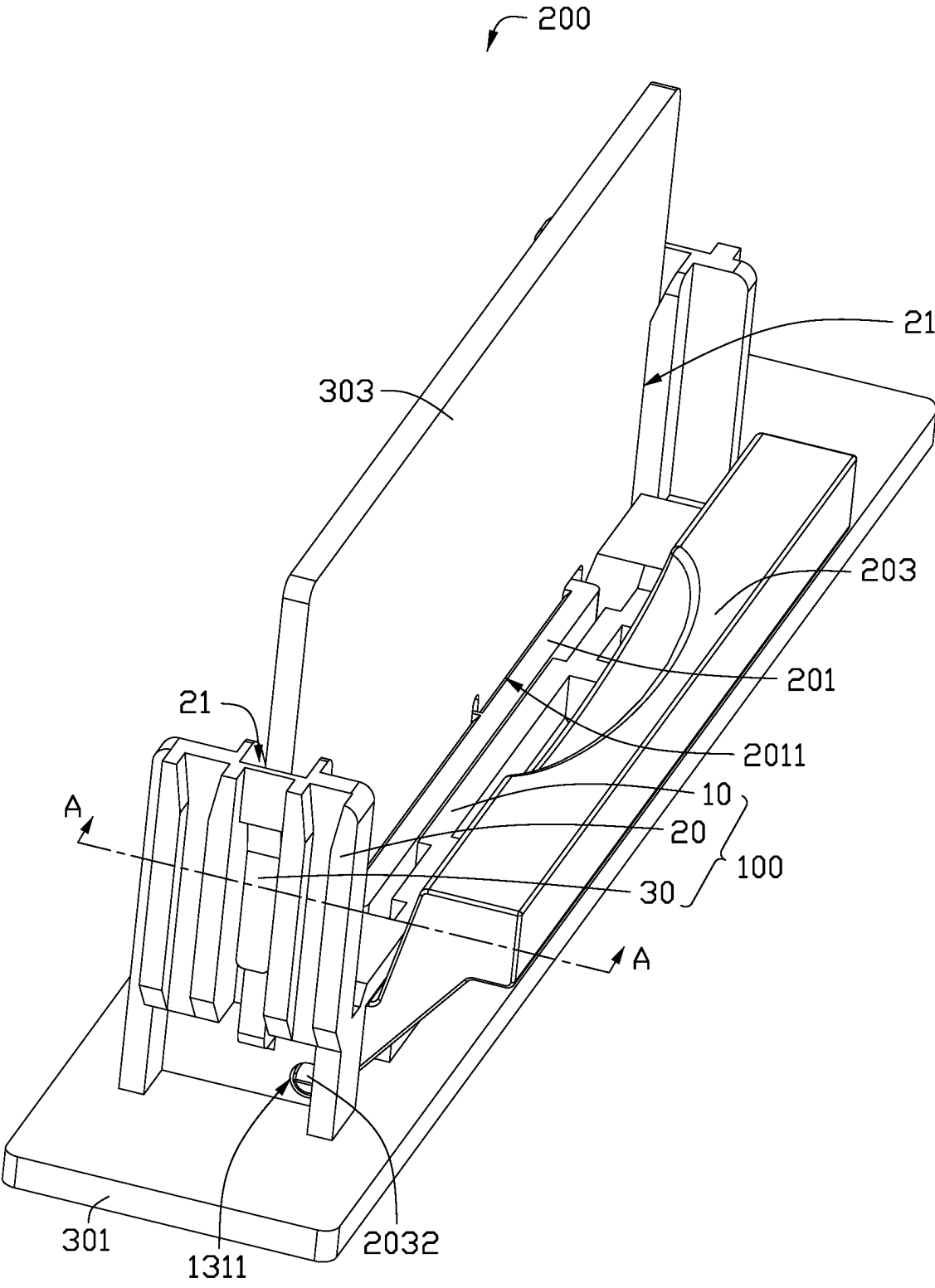


FIG. 1

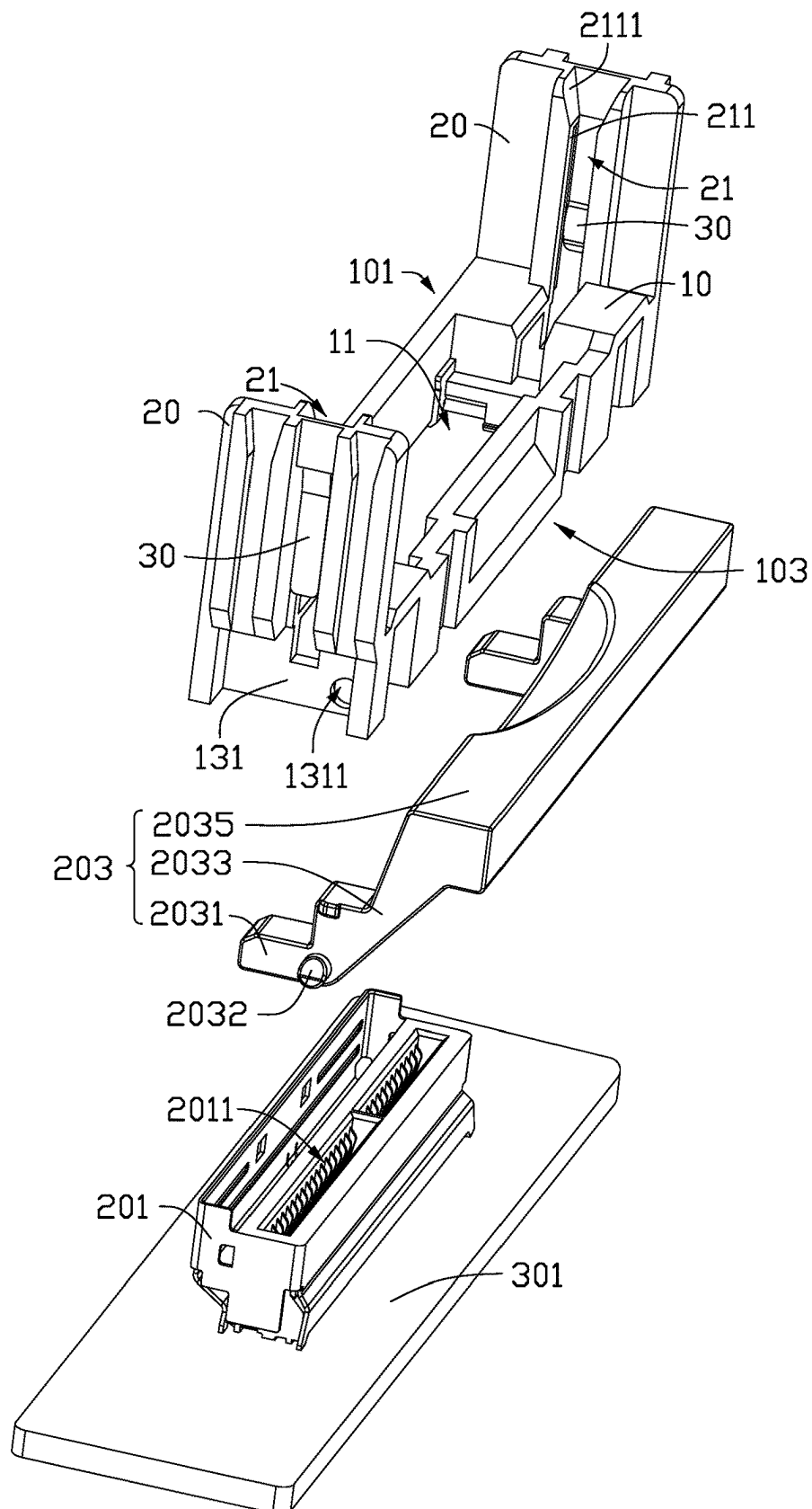


FIG. 2

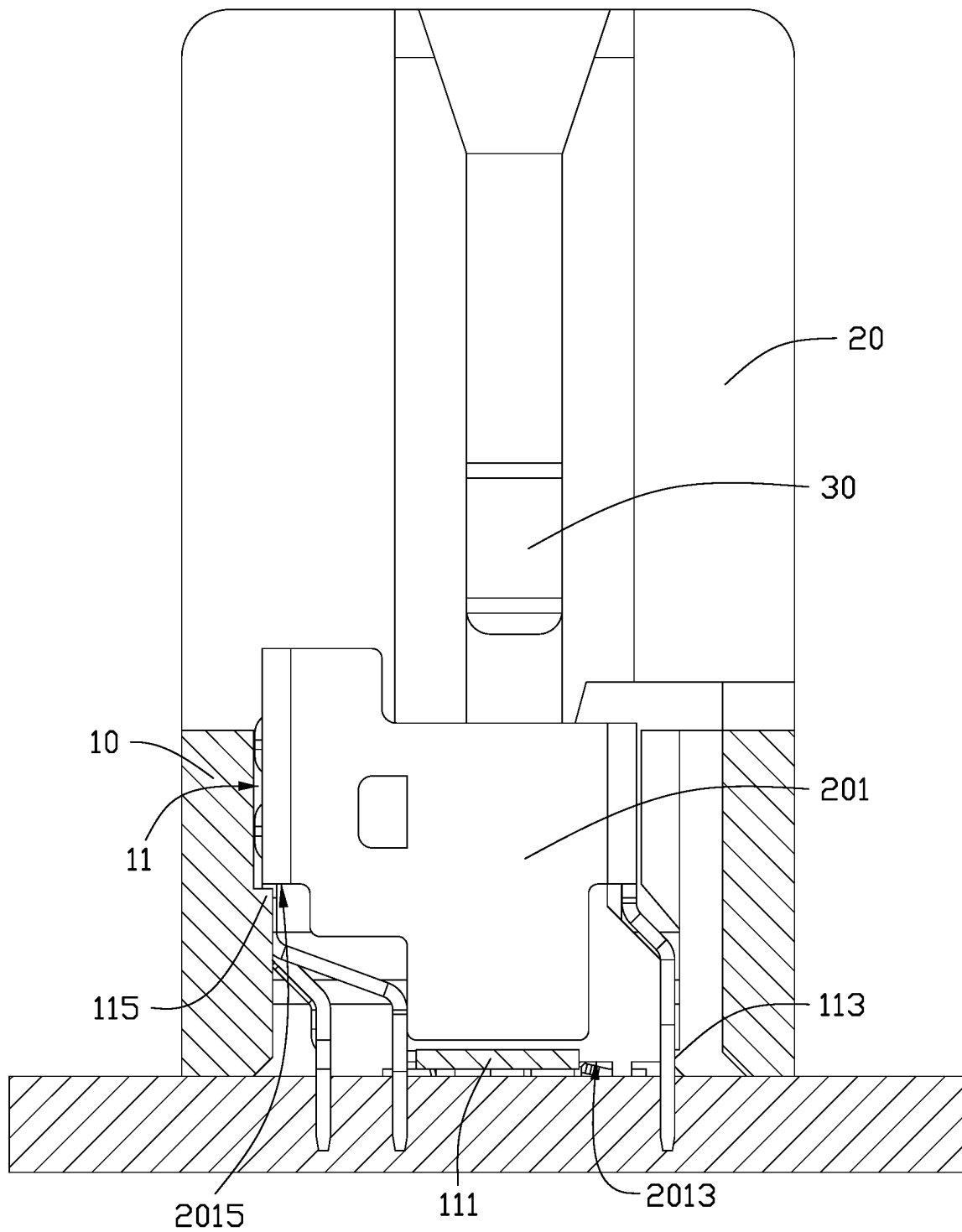


FIG. 3

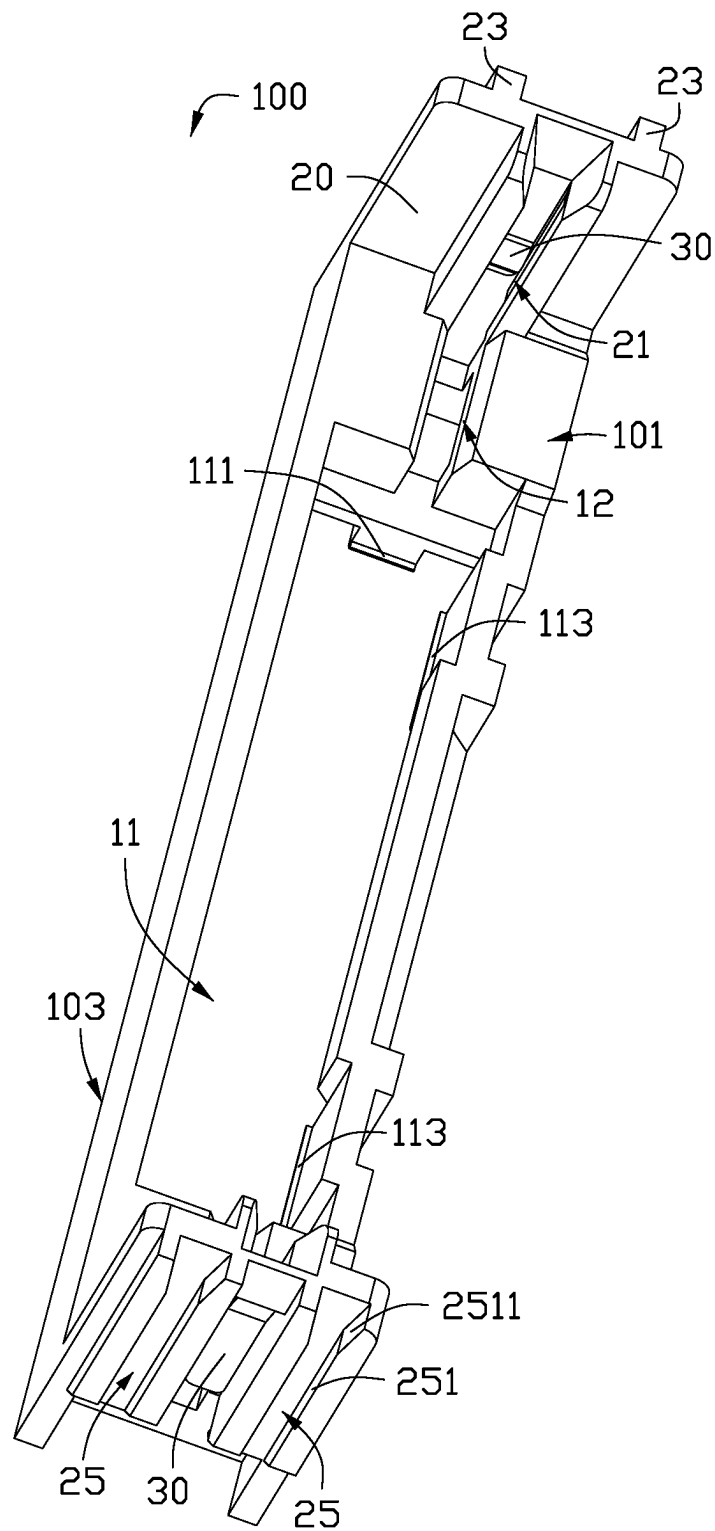


FIG. 4

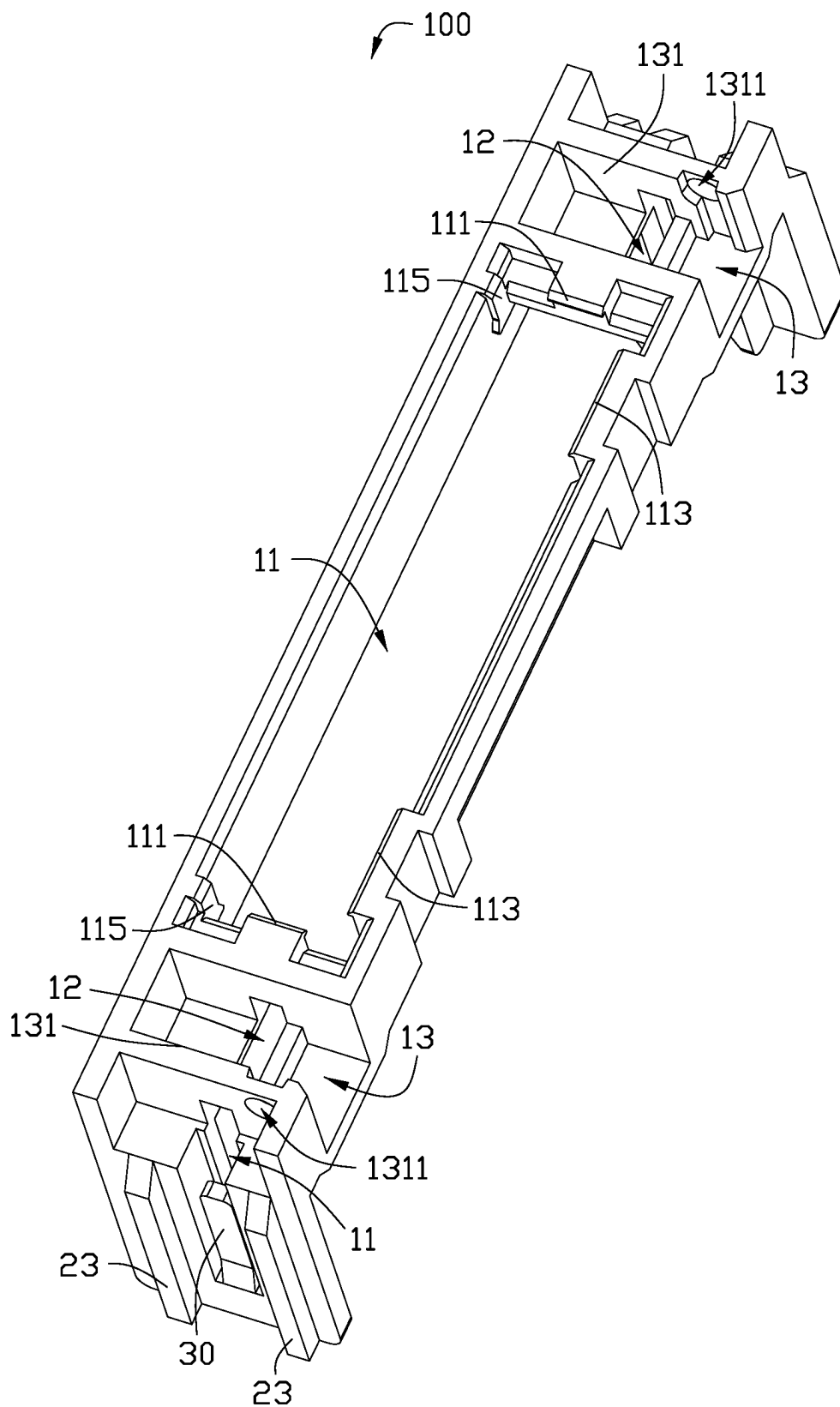


FIG. 5

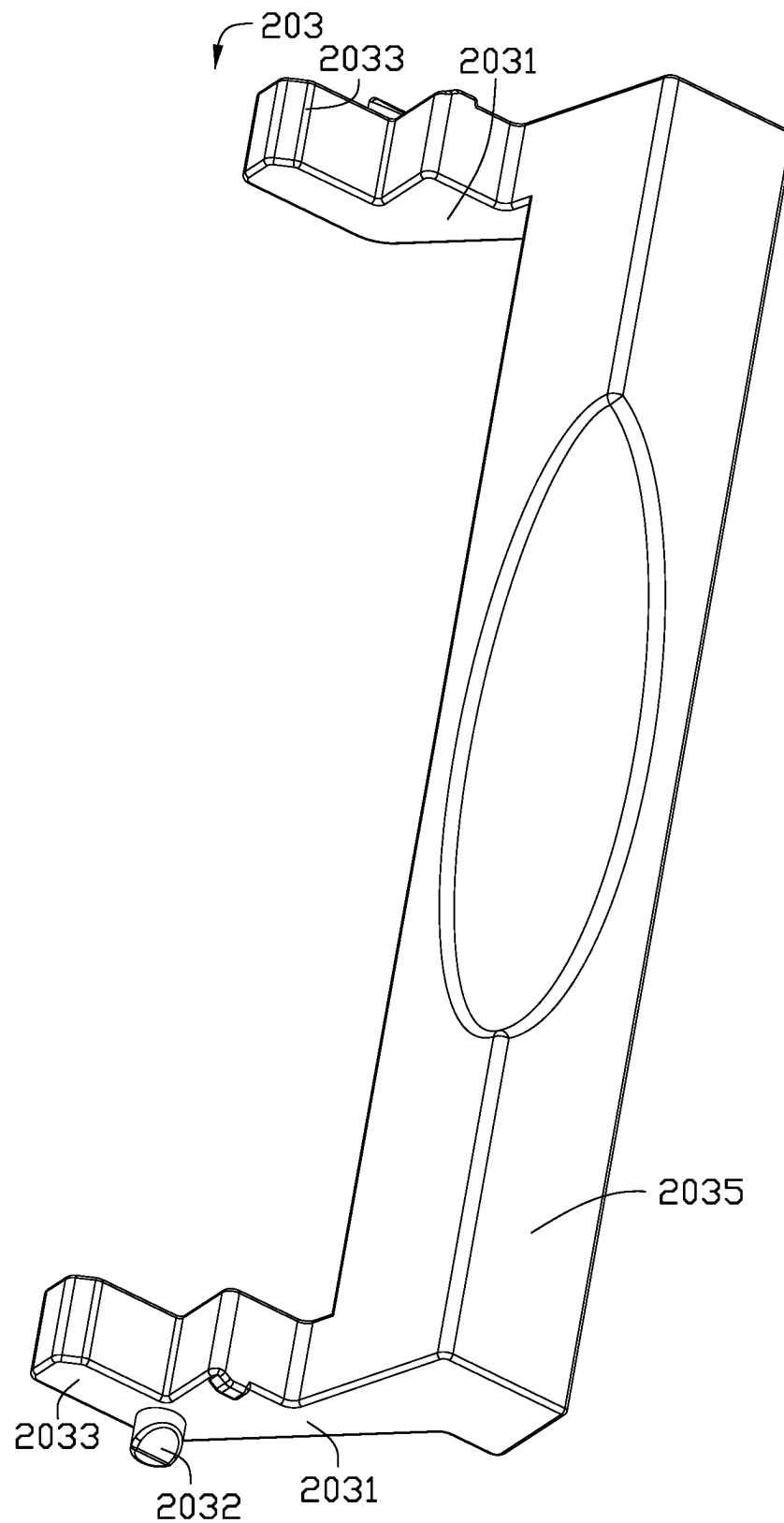


FIG. 6

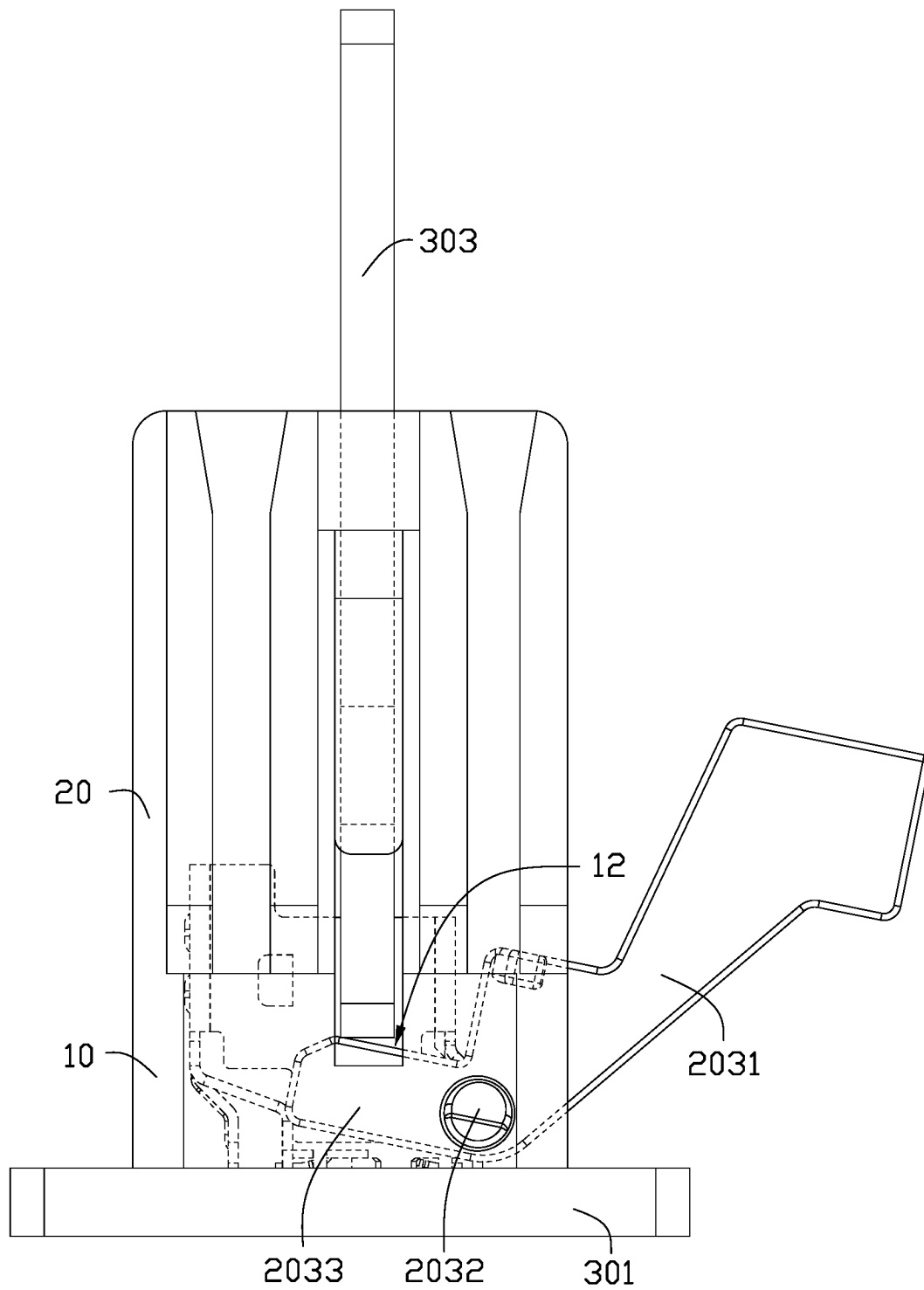


FIG. 7

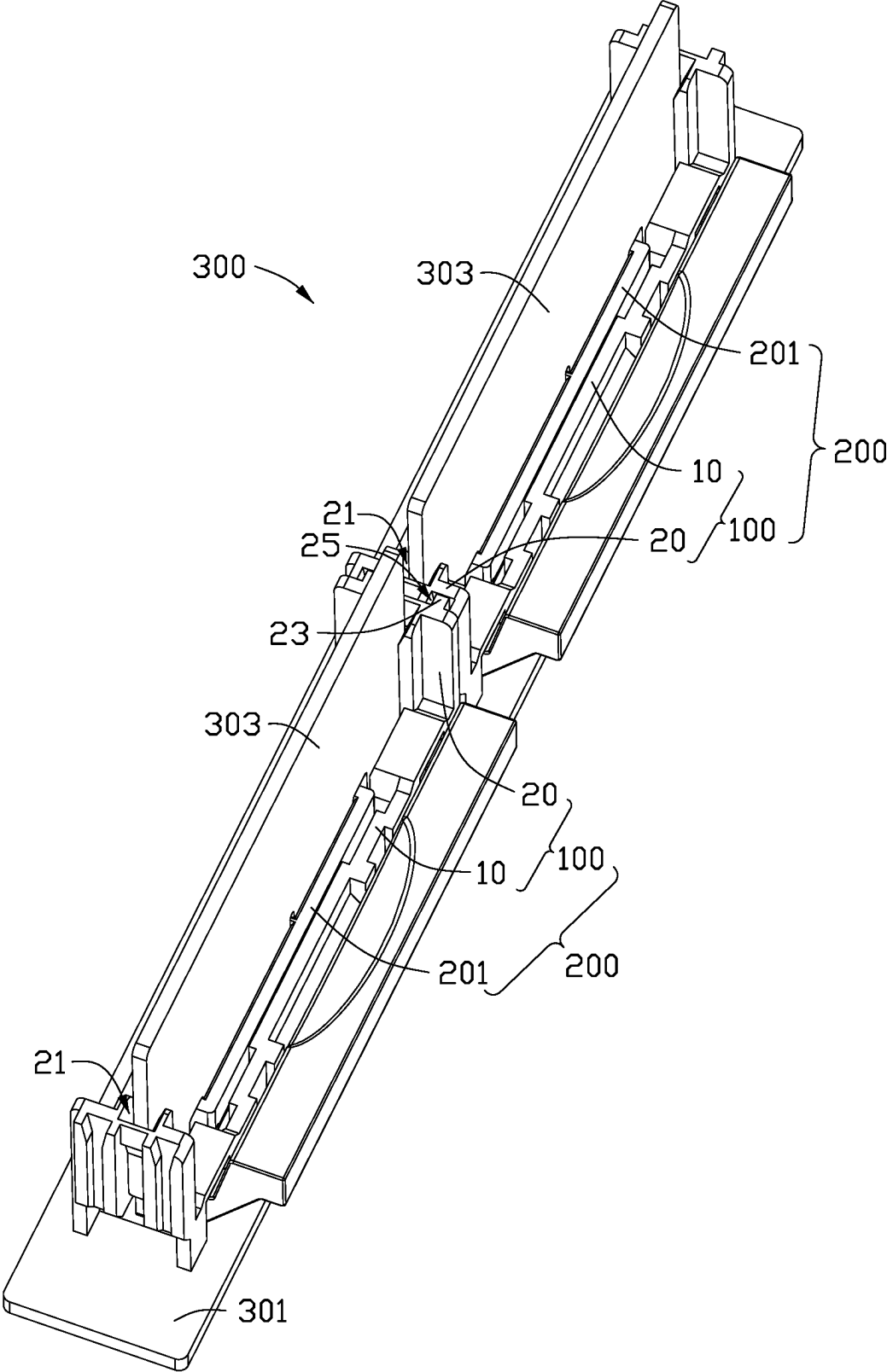


FIG. 8

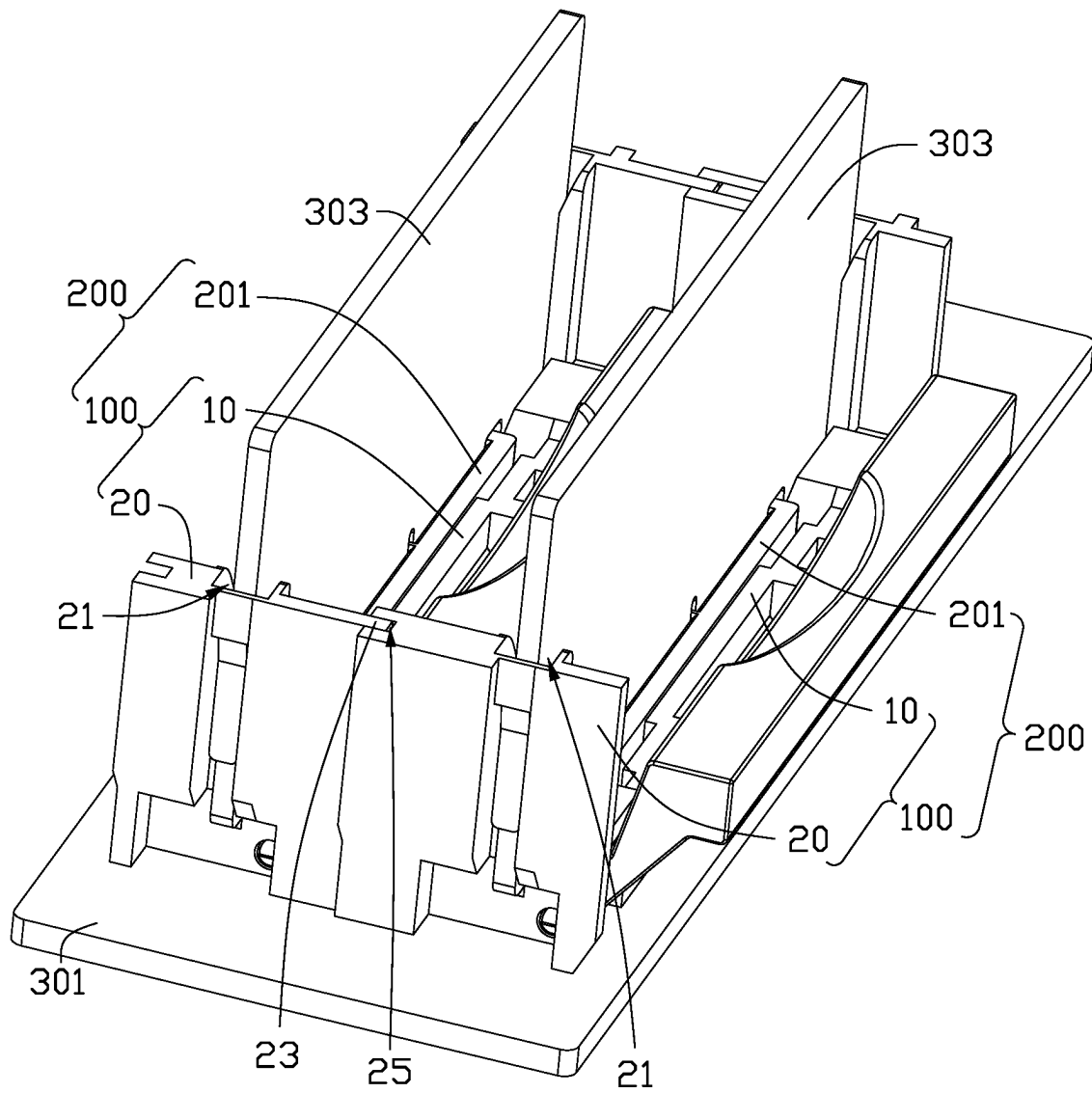


FIG. 9

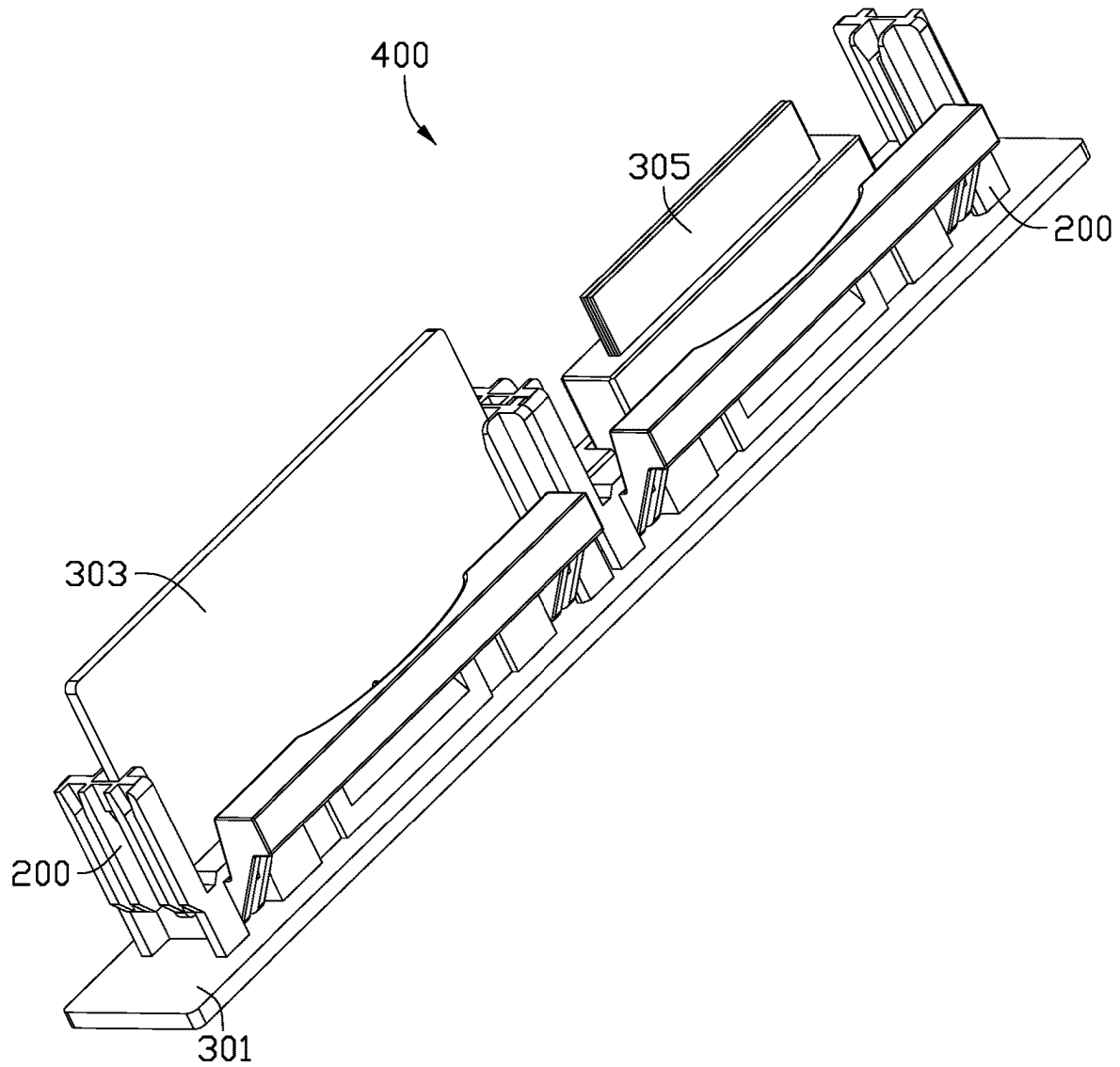


FIG. 10

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**ELECTRICAL CONNECTION BASE,
ELECTRICAL CONNECTION BASE
ASSEMBLY WITH ENHANCED
LOAD-BEARING AND SUPPORTIVE
ABILITY, AND ELECTRICAL CONNECTOR**

FIELD

The subject matter herein generally relates to housings.

BACKGROUND

As the functions and capabilities of external circuit boards such as expansion cards increase, components such as weighty heat-dispersing structures may also be installed on the expansion cards, resulting in an increase of loading on the expansion card. The expansion card is supported by the connector, but the load-bearing part of the expansion card is not supported, which may cause deformation and damage to the expansion card.

BRIEF DESCRIPTION OF THE DRAWINGS

Many aspects of the disclosure could be better understood with reference to the following drawings. The components in the drawings are not necessarily drawn to scale, the emphasis instead being placed upon clearly illustrating the principles of the disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

FIG. 1 is an isometric, assembled view of an electrical connection base connecting a first circuit board and a second circuit board provided by an embodiment of the present application.

FIG. 2 is an isometric, exploded view of the electrical connection base shown in FIG. 1.

FIG. 3 is a sectional view along line A-A of FIG. 1.

FIG. 4 is an isometric view of a supporting element in the electrical connection base shown in FIG. 1.

FIG. 5 is an isometric view of the supporting element shown in FIG. 4 from another perspective.

FIG. 6 is an isometric, assembled view of an operating element in the electrical connection base shown in FIG. 1.

FIG. 7 is an isometric view of the electrical connection base shown in FIG. 1 wherein the electrical connection base is in a state ready for installation or removal of a second circuit board.

FIG. 8 is an isometric view of an electrical connection base assembly provided by an embodiment of the present application.

FIG. 9 is an isometric view of the electrical connection base assembly shown in FIG. 8 in another embodiment.

FIG. 10 is an isometric view of an electrical connector provided by an embodiment of the present application.

DETAILED DESCRIPTION

It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein could be practiced without these specific details. In other instances, methods, procedures, and components have not been described in

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detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale and the proportions of certain parts have been exaggerated to better illustrate details and features of the present disclosure.

The present disclosure, including the accompanying drawings, is illustrated by way of examples and not by way of limitation. Several definitions that apply throughout this disclosure will now be presented. It should be noted that references to “an” or “one” embodiment in this disclosure are not necessarily to the same embodiment, and such references mean “at least one.”

The term “coupled” is defined as connected, whether directly or indirectly through intervening components, and is not necessarily limited to physical connections. The connection could be such that the objects are permanently connected or releasably connected. The term “comprising” means “including, but not necessarily limited to”; it specifically indicates open-ended inclusion or membership in a so-described combination, group, series, and the like.

Without a given definition otherwise, all terms used have the same meaning as commonly understood by those skilled in the art. The terms used herein in the description of the present disclosure are for the purpose of describing specific embodiments only, and are not intended to limit the present disclosure.

As shown in FIG. 1 and FIG. 2, in an embodiment, an electrical connection base 200 is illustrated. The electrical connection base 200 includes a pedestal 201 and a supporting element 100. The pedestal 201 is positioned on a first circuit board 301 and is electrically connected to the first circuit board 301. A socket 2011 is defined on the pedestal 201, and the socket 2011 can be configured to insert a second circuit board 303 for electrical connection to the first circuit board 301.

In an embodiment, the first circuit board 301 is a main board, and the second circuit board 303 is an expansion card. In another embodiment, the first circuit board 301 can also be a sub-circuit board electrically connected to the main board, and the second circuit board 303 can also be a cable or another component electrically switched by the electrical connection base 200.

The supporting element 100 includes an installation portion 10 and a support portion 20. Installation portion 10 is configured for an installation to an exterior of the pedestal 201. The installation portion 10 includes opposing first and second sides 101 and 103. The second side 103 of the installation portion 10 faces the first circuit board 301. A through hole 11 is defined in the installation portion 10. The through hole 11 penetrates the first side 101 and the second side 103. The installation portion 10 moves toward the first circuit board 301 and is set outside the pedestal 201, so that the pedestal 201 extends through the through hole 11. The socket 2011 is exposed from the first side 101 of the installation portion 10. The pedestal 201 may be entirely located in the through hole 11 or may partially protrude outside the through hole 11. The support portion 20 is set on the first side 101 of the installation portion 10. A first slot 21 is defined on the side of the support portion 20 facing the through hole 11, and the first slot 21 penetrates an end of the support portion 20 away from the installation portion 10.

When the second circuit board 303 is plugged into the socket 2011, the wall of the second circuit board 303 enters the first slot 21, a groove wall 211 of the first slot 21 contacts

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the second circuit board **303**, and the support portion **20** supports the second circuit board **303** against strain and deformation.

In an embodiment, in order to facilitate entry of the wall of the second circuit board **303** in the first slot **21**, a first guide surface **2111** is provided on the two opposite groove walls **211** of the first slot **21** facing away from the installation portion **10**. The first guide surface **2111** is an inclined surface. In another embodiment, the first guide surface **2111** may also be a curved surface.

In an embodiment, the supporting element **100** further includes a resistance portion **30**. The resistance portion **30** is positioned in the support portion **20** and in the first slot **21**. The resistance portion **30** constrains and supports the part of the second circuit board **303** which is inserted into the first slot **21**.

The resistance portion **30** is made of elastically deformable materials such as plastic. When second circuit boards **303** have different widths, the walls of the second circuit board **303** entering the first slot **21** bear against and compress the resistance portion **30**, so that the resistance portion **30** deforms to closely fit second circuit boards **303** of different sizes.

One end of the resistance portion **30** is connected to the support rod **20**, and the other end of the resistance portion **30** is a free end. The free end of the resistance portion **30** abuts on the second circuit board **303**, which facilitates elastic deformation of the resistance portion **30** by different widths of second circuit boards **303**.

In another embodiment, the resistance portion **30** may also be other elastic structures such as a spring in the form of a disk positioned in the first slot **21**, and the resistance portion **30** will thereby bear on the second circuit board **303**.

In an embodiment, there are two support portions **20**. A support portion **20** is located on each side of the through hole **11**. Opposite walls of the second circuit board **303** are inserted into the first slots **21** on the support portion **20**, and the two support portions **20** provide support. In another embodiment, there may be only one support portion **20**.

As shown in FIG. 3, FIG. 4, and FIG. 5, in an embodiment, a number of buckle parts protrude from the hole wall of the through hole **11**, and the buckle parts can be buckled on a side of the pedestal **201** facing the first circuit board **301**. That is, the buckle parts are located on a side of the pedestal **201** away from the first side **101** of the installation portion **10**. The buckle parts include two first buckles **111**, two second buckles **113**, and two third buckles **115**. The two first buckles **111** are positioned on opposite walls of the narrow side of the through hole **11** close to the support portions **20**, and the second buckle **113** and the third buckle **115** are positioned on the other two walls (the long walls) of the through hole **11**. The two second buckles **113** are positioned on the same hole wall of the through hole **11**, and the two third buckles **115** are positioned on the same hole wall of the through hole **11**. There is a gap between a bottom wall **2013** of the pedestal **201** facing the first circuit board **301** and the first circuit board **301**, and the first buckle **111** extends through the gap and abuts the bottom wall **2013**.

There is a gap between a bottom wall **2013** of the base **201** facing the first circuit board **301** and the first circuit board **301**, and the second engaging portion **113** extends through the gap and abuts the bottom wall **2013**. There is a gap between bottom wall **2015** of the base **201** facing the first circuit board **301** and the first circuit board **301**, and the third engaging portion **115** extends through the gap and abuts the

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bottom wall **2015**. The bottom wall **2013** and the bottom wall **2015** are on different planes.

In another embodiment, the buckle parts may be provided at other positions on the hole wall of the through hole **11**, and the number of engaging portions may be one or more.

In another embodiment, the engaging portions may be omitted. The pedestal **201** and the hole wall of the through hole **11** have an interference fit or transition fit, and the hole wall of the through hole **11** presses against the outer peripheral wall of the pedestal **201**. The pedestal **201** and the installation portion **10** are held firmly in place by the pressing force and friction between the hole wall of the through hole **11** and the pedestal **201**.

In an embodiment, the supporting element **100** is an injection molding and an integrated structure. In the process of installing the supporting element **100** outside the pedestal **201**, the buckle parts are elastically deformed and buckled outside the pedestal **201**.

In another embodiment, the supporting element **100** may also be a composite structure of plastic and metal.

As shown in FIG. 6, and FIG. 7, in order to facilitate disassembly of the second circuit board **303** from the pedestal **201**, the electrical connection base **200** includes an operating element **203**. A second slot **12** is defined on the first side **101** of the installation portion **10**. A third slot **13** is defined on the second side **103** of the installation portion **10**, as shown in FIG. 5. The second slot **12** communicates with the third slot **13** and is located between the first slot **21** and the through hole **11**. The second slot **12** also communicates with the first slot **21** and the through hole **11**. The second circuit board **303** is substantially a rectangular plate. The side of the second circuit board **303** facing the first circuit board **301** can enter the second slot **12**.

The operating element **203** includes a connecting portion **2031** and a contacting portion **2033**. The connecting portion **2031** is positioned in the third slot **13** and partially protrudes out of the installation portion **10**, it can be rotated. The connecting portion **2031** rotates and drives the contacting portion **2033** to enter or exit the second slot **12**. When the contacting portion **2033** enters the second slot **12**, the contacting portion **2033** contacts the second circuit board **303**, and as the connecting portion **2031** rotates relative to the installation portion **10**, the contacting portion **2033** abuts the second circuit board **303** and moves the second circuit board **303** along the first slot **21**.

When the second circuit board **303** is inserted into the socket **2011**, the second circuit board **303** can move toward the first circuit board **301** along the first slot **21**. The second circuit board **303** presses against the contacting portion **2033** and drives the contacting portion **2033** to move until the contacting portion **2033** exits the second slot **12**. The second circuit board **303** is thereby plugged into the socket **2011**. The operating element **203** is relatively fixed under the second circuit board **303**.

When the second circuit board **303** is disassembled, an external force is applied to the operating element **203**, the connecting portion **2031** rotates relative to the installation portion **10**, the contacting portion **2033** enters the second slot **12**, and forces the second circuit board **303** away from the first circuit board **301** until the two are separated.

A mounting hole **1311** is defined in the groove wall **131** of the third slot **13** away from the through hole **11**. The connecting portion **2031** comprises a shaft **2032**. The shaft **2032** is rotatable in the mounting hole **1311**.

In an embodiment, there are two second slots **12** and two third slots **13**, and there are two connecting portions **2031** and two contacting portions **2033**. The operating element

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203 includes an operating portion **2035**. One end of the operating portion **2035** is connected to an end of the connecting portion **2031** facing away from the contacting portion **2033**, and the other end of the operating portion **2035** extends to one side of the connecting portion **2031**. The operating portion **2035** connects the two connecting portions **2031**. The external force for dismounting the second circuit board **303** is applied to the operating portion **2035**, the leverage makes it easy to dismount the second circuit board **303**.

In another embodiment, the number of second slots **12**, third slots **13**, and connecting portion **2031** and contacting portion **2033** of the operating element **203** may be one of each.

In another embodiment, the operating element **203** may be omitted, and the external force may be directly applied to the part of the connecting Portion **2031** protruding outside the installation portion **10**.

In another embodiment, when the second circuit board **303** is an unusual shape, for example, the second circuit board **303** includes three sections with different heights, and the three sections can extend into the corresponding first slot **21**, second slot **12** and through hole **11**, the first slot **21** and through hole **11** can be disconnected from the second slot **12**.

As shown in FIG. 4 and FIG. 8, an embodiment of the present application provides an electrical connection base assembly **300**. The electrical connection base assembly **300** includes a plurality of electrical connection bases **200**. Several electrical connection bases **200** are positioned on the first circuit board **301** along the connection direction of the two support portions **20** of the supporting element **100**. One of the two support portions **20** of the supporting element **100** comprises a bulge card **23**, and a holding slot **25** is defined on the other of the two support portions **20**. The bulge card **23** and the holding slot **25** are located on opposite sides of the two support portions **20**.

One second circuit board **303** can be plugged into each electrical connection base **200**. Among adjacent electrical connection bases **200**, the bulge card **23** of one electrical connection base **200** can be clamped into the holding slot **25** of the other electrical connection base **200**. The adjacent supporting elements **100** support each other, which improves the stability of the supporting element **100** and further improves the support stability of the supporting element **100** to the second circuit board **303**.

In an embodiment, the number of bulge cards **23** and holding slots **25** of each supporting element **100** is two. In another embodiment, the number of bulge cards **23** and holding slots **25** of the supporting element **100** may be one, three, four, or other number.

In order to facilitate entry of the bulge card **23** into the holding slot **25**, a second guide surface **2511** is defined on the groove wall **251** of the holding slot **25**.

In another embodiment, the electrical connection bases **200** may be positioned on the first circuit board **301** along a direction perpendicular to the connecting line between the two support portions **20** of the supporting element **100**, as shown in FIG. 9. Accordingly, the bulge card **23** and the holding slot **25** are positioned on opposite sides of the same support portion **20** along the arrangement direction of the electrical connection bases **200**.

The socket **2011** is not limited to the second circuit board **303** plugging into it. For example, in another embodiment, as shown in FIG. 10, an embodiment of the present application provides an electrical connector **400**. The electrical connector **400** includes two electrical connection bases **200**, two first circuit boards **301**, one second circuit board **303**,

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and one cable **305**. Two electrical connection bases **200** are positioned on the first circuit boards **301**, and one of the electrical connection bases **200** is plugged into a cable **305**. The supporting element **100** is set outside the pedestal **201** and does not affect the cable **305** plugged into the pedestal **201**.

In another embodiment, the electrical connector **400** includes one electrical connection base **200**, one first circuit board **301**, and one second circuit board **303** or cable **305**. The second circuit board **303** or cable **305** is plugged into the electrical connection base **200**.

In the electrical connection base **200**, the supporting element **100** is set outside the pedestal **201**, and the first slot **21** is defined on the supporting element **100**, so that the part of the element plugged into the socket **2011** and outside the pedestal **201** is inserted into the first slot **21**. The groove wall **211** of the first slot **21** is in contact with the element, so that the support portion **20** supports the element and prevents deformation of the element.

The embodiments shown and described above are only examples. Even though numerous characteristics and advantages of the present technology have been set forth in the foregoing description, together with details of the structure and function of the present disclosure, the disclosure is illustrative only, and changes may be made in the detail, including in matters of shape, size and arrangement of the parts within the principles of the present disclosure, up to and including the full extent established by the broad general meaning of the terms used in the claims.

What is claimed is:

1. An electrical connection base comprising:

a pedestal configured to be positioned on a first circuit board; a socket defined on the pedestal and configured to receive a second circuit board for establishing electrical connection between the second circuit board and the first circuit board; and

a supporting element comprising:

an installation portion, a through hole defined on the installation portion, wherein the installation portion is installed to an exterior of the pedestal, and the pedestal extends through the through hole, and the socket is exposed from a first side of the installation portion; and

at least one support portion positioned on a first side of the installation portion and outside of the through hole, wherein a first slot is defined on a side of the support portion facing the through hole, and the first slot penetrates an end of the support portion away from the installation portion;

when the second circuit board is plugged into the socket, the wall of the second circuit board is in the first slot, a groove wall of the first slot contacts the second circuit board, and the support portion supports the second circuit board.

2. The electrical connection base of claim 1, wherein:

a buckle part protrudes from a hole wall of the through hole, and the buckle part is located on a side of the pedestal away from the first side of the installation portion.

3. The electrical connection base of claim 2, wherein:

a gap between a bottom wall of the pedestal and the first circuit board is defined, and the buckle part extends through the gap and abuts on the bottom wall.

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4. The electrical connection base of claim 1, wherein: the pedestal and a hole wall of the through hole are in interference fit or transition fit, and the hole wall of the through hole presses an outer peripheral wall of the pedestal.
5. The electrical connection base of claim 1, wherein: a second slot is defined on the first side of the installation portion, the electrical connection base further comprises an operating element, the operating element comprises at least one connecting portion and at least one contacting portion connected to a respective one of the at least one connecting portion;
- the connecting portion is rotatably positioned on the installation portion and partially protrudes from the installation portion; the connecting portion rotates and drives the contacting portion to enter or exit the second slot.
6. The electrical connection base of claim 5, wherein: a third slot is defined on a second side of the installation portion, and the first side and the second side that are opposite sides of the installation portion;
- the second slot communicates with the third slot and is located between the first slot and the through hole; the connecting portion is rotatably positioned in the third slot.
7. The electrical connection base of claim 6, wherein: the second slot communicates with the first slot and the through hole.
8. The electrical connection base of claim 6, wherein: a mounting hole is defined in the groove wall of the third slot;
- the connecting portion comprises a shaft, and the shaft is rotatably in the mounting hole.
9. The electrical connection base of claim 8, wherein: the mounting hole is located in the groove wall of the third slot away from the through hole.
10. The electrical connection base of claim 5, wherein: the operating element further comprises an operating portion, one end of the operating portion is connected to an end of the connecting portion facing away from the contacting portion, and the other end of the operating portion extends to one side of the connecting portion.
11. The electrical connection base of claim 10, wherein: there are two support portions, the two support portions are located on opposite sides of the through hole.
12. The electrical connection base of claim 11 comprising: two of the connecting portions and two of the contacting portions, the first slot is defined on each of the support portions, the two connecting portions are positioned at intervals on the installation portion, and the operating portion is connected to the two connecting portions.
13. The electrical connection base of claim 1, wherein: when the wall of the second circuit board is in the first slot; the resistance portion is positioned in the support portion and in the first slot, the resistance portion resists the second circuit board plugged into the first slot.
14. The electrical connection base of claim 13, wherein: the resistance portion is made of elastically deformable materials.
15. The electrical connection base of claim 14, wherein: one end of the resistance portion is connected to the support rod, and the other end of the resistance portion is a free end.

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16. The electrical connection base of claim 1, wherein: a first guide surface is provided on the two opposite groove walls of the first slot facing away from the installation portion.
17. The electrical connection base of claim 1 comprising: two of the support portions, the two support portions are located on opposite sides of the through hole;
- one of the two support portions comprising a bulge card; and a holding slot is defined in the other of the two support portions;
- the bulge card and the holding slot are located on opposite sides of the two support portions.
18. The electrical connection base of claim 17, wherein: a second guide surface is defined on the groove wall of the holding slot, and the second guide surface is located at the end of the support portion away from the installation portion.
19. An electrical connection base assembly comprising: a plurality of electrical connection bases, each of the plurality of electrical connection bases comprising: a pedestal a socket defined on the pedestal and configured to receive a second circuit board to establish electrical connection between the second circuit board and a first circuit board; and
- a supporting element comprising: an installation portion, a through hole defined on the installation portion, wherein the installation portion is installed to an exterior of the pedestal, and the pedestal extends through the through hole, and the socket is exposed from a first side of the installation portion; and
- two of the support portions positioned at intervals on the first side of the installation portion, and located on both sides of the through hole, wherein when the second circuit board is plugged into the socket, the wall of the second circuit board is in the first slot, a groove wall of the first slot contacts the second circuit board, and the support portion supports the second circuit board;
- a first slot defined on a side of the support portion facing the through hole, and the first slot penetrating an end of the support portion away from the installation portion; wherein:
- the plurality of the electrical connection bases are positioned in sequence, one of the two support portions of the supporting element comprises a bulge card, and a holding slot is defined on another of the two support portions, and the bulge card and the holding slot are located on opposite sides of the two support portions.
20. An electrical connector comprising: a first circuit board;
- a second circuit board or a cable; and
- an electrical connection base comprising: a pedestal positioned on the first circuit board and electrically connected to the first circuit board; and a socket defined on the pedestal; and
- a supporting element comprising: an installation portion, a through hole defined on the installation portion, wherein the installation portion is installed to an exterior of the pedestal, and the pedestal extends through the through hole, and the socket is exposed from a first side of the installation portion, and is configured to receive the second circuit board or the cable to establish electrical connection between the second circuit board and the first circuit board; and

at least one support portion positioned on the first side
of the installation portion, a first slot defined on a
side of the support portion facing the through hole,
and the first slot penetrating an end of the support
portion away from the installation portion, wherein 5
when the second circuit board is plugged into the
socket, the wall of the second circuit board is in the
first slot, a groove wall of the first slot contacts the
second circuit board, and the support portion sup-
ports the second circuit board; and 10
the second circuit board or the cable is in the first slot.

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